Guo Jingxiang

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Education

National University of Singapore **May. 2026 – Present**
Doctor of Philosophy (PhD) *Singapore*
- Advisor: Prof. Shuicheng Yan; Co-advisor: Prof. Jianshu Zhou

National University of Singapore **Aug. 2025 – May. 2026**
Master of Computing in Artificial Intelligence *Singapore*

Shanghai Jiao Tong University **May. 2025 – Aug. 2025**
Visiting Student *Shanghai, China*

Harbin Institute of Technology, Shenzhen **Sep. 2021 – Jul. 2025**
Bachelor of Engineering in Automation *Shenzhen, China*

National University of Singapore **Jul. 2024 – May. 2025**
NGNE Program Exchange Student *Singapore*

Research Experience

Learning and Vision Lab (LV-Lab), NUS **May. 2026 – Present**
PhD Student *Singapore*

CLeAR lab, NUS **Aug. 2025 – May. 2026**
Research Intern *Singapore*
- Developing a tactile foundation model for heterogeneous sensors.

ScaleLab, SJTU **May. 2025 – Aug. 2025**
Research Intern *Shanghai, China*
- Developed a vision–tactile data collection pipeline to support a benchmark towards foundation visuo-tactile policy learning.

LinS Lab, NUS **Jul. 2024 – May. 2025**
Research Assistant *Singapore*
- Developed a user-friendly teleoperation data collection system, a cross-embodiment dexterous grasping method, and a world model-driven Vision-Language-Object-Action model.

RLGroup Lab, HITsz **Oct. 2022 – Jun. 2024**
Research Assistant *Shenzhen, China*
- Investigated Adaptive Logarithmic Basis Policy Gradient algorithms and proposed a Multi-agent Reinforcement Learning method to improve locomotion learning for a single quadruped robot.

Employment Experience

Tencent **Dec. 2025 – Apr. 2026**
Data Scientist Intern *Singapore*
- Built a player base agent for a game-world simulation, using a three-stage pipeline (domain vocabulary expansion, action-history SFT for action prediction, and GRPO on full logs) to support live-ops outcome forecasting and player churn analysis.

Autolife Robotics **Aug. 2025 – Nov. 2025**
Robot Software System Intern *Singapore*
- Working on wheel-based humanoid software system development and implementation.

Horizon Robotics **Jun. 2025 – Aug. 2025**
Cloud Platform Intern *Shanghai, China*
- Participated in conducting a workshop on benchmarking generalizable bimanual manipulation using the RoboTwin platform.

Publications

1. **Jingxiang Guo**, Haoyu Zhao, Xidong Zhang, Yuhang Zheng, Hao Wu, Chen Gao, Jianshu Zhou, Shuicheng Yan, Weihao Yuan, "**UMI-WM: Interactive World Modeling for Bimanual Manipulation from Wrist Cameras**". In submission. Project Page | arXiv

2. Haonan Chen*, **Jingxiang Guo***, Bangjun Wang, Tianrui Zhang, Xuchuan Huang, Boren Zheng, Yiwen Hou, Chenrui Tie, Jiajun Deng, Lin Shao, "**Goal-VLA: Image-Generative VLMs as Object-Centric World Models Empowering Zero-shot Robot Manipulation**". Accepted to ICRA 2026.

3. **Jingxiang Guo***, Jiayu Luo*, Zhenyu Wei*, Yiwen Hou, Zhixuan Xu, Xiaoyi Lin, Chongkai Gao, Lin Shao, "**TelePreview: A User-Friendly Teleoperation System with Virtual Arm Assistance for Enhanced Effectiveness**". Accepted to ICRA 2025 @ Human-Centric Multilateral Teleoperation, **Best Paper Award**.

4. Zhenyu Wei*, Zhixuan Xu*, **Jingxiang Guo**, Yiwen Hou, Chongkai Gao, Zhehao Cai, Jiayu Luo, Lin Shao, "$\mathcal{D}(\mathcal{R}, \mathcal{O})$ **Grasp: A Unified Representation of Robot and Object Interaction for Cross-Embodiment Dexterous Grasping**". Accepted to ICRA 2025 (**Best Paper Award on Robot Manipulation and Locomotion**); CoRL 2024 @ MAPoDeL, **Best Robotics Paper Award** & **Oral Presentation**; CoRL 2024 @ LFDM, **Spotlight Presentation**.

5. Chenrui Tie, Shengxiang Sun, Jinxuan Zhu, Yiwei Liu, **Jingxiang Guo**, Yue Hu, Haonan Chen, Junting Chen, Ruihai Wu, Lin Shao, "**Manual2Skill: Learning to Read Manuals and Acquire Robotic Skills for Furniture Assembly Using Vision-Language Models**". Accepted to RSS 2025; CVPR 2025 @ 3D Vision Language Models, **Oral Presentation**.

6. Jinxuan Zhu*, Chenrui Tie*, Xinyi Cao*, Yuran Wang, **Jingxiang Guo**, Zixuan Chen, Haonan Chen, Junting Chen, Yangyu Xiao, Ruihai Wu, Lin Shao, "**AdaptPNP: Integrating Prehensile and Non-Prehensile Skills for Adaptive Robotic Manipulation**". Accepted to ICRA 2026.

7. Jinxuan Zhu*, Zihao Yan*, Yangyu Xiao, **Jingxiang Guo**, Chenrui Tie, Xinyi Cao, Yuhang Zheng, Lin Shao, "**ShapeForce: Low-Cost Soft Robotic Wrist for Contact-Rich Manipulation**". Accepted to ICRA 2026; IROS 2025 @ Contact and Impact-aware Manipulation, **Best Demo Award Finalist**.

8. Haonan Chen, Junxiao Li, Chongkai Gao, Zhixuan Xu, Chenting Wang, Yiwen Hou, **Jingxiang Guo**, Shensi Xu, Jiaqi Huang, Weidong Wang, Lin Shao, "**MetaFold: A Closed-loop Pipeline for Universal Clothing Folding via End-to-end Point Cloud Trajectory Generation**". Accepted to IROS 2025 (**Oral Presentation**).

9. Lixin Xu, Zixuan Liu, Zhewei Gui, **Jingxiang Guo**, Zeyu Jiang, Zhixuan Xu, Chongkai Gao, Lin Shao, "**DexSinGrasp: Learning a Unified Policy for Dexterous Object Singulation and Grasping in Cluttered Environments**". Accepted to RA-L; ICRA 2025 @ Handy Moves, **Spotlight Presentation**.

10. Congsheng Xu*, **Jingxiang Guo***, Baijun Chen*, Liuhaichen Yang, Zhen Zou, Yuzhang Li, Jieji Ren, Yiming Wang, Yichao Yan, Yao Mu, Xiaokang Yang, "**InsertScale: A Benchmark towards Foundation Visuo-Tactile Policy Scalable Learning for Insertion Task**". IROS 2025 (Late Breaking Results); IROS 2025 @ Learning from Teleoperation, **Oral Presentation**.

11. Qi Liu*, **Jingxiang Guo***, Sixu Lin, Shuaikang Ma, Jinxuan Zhu, Yanjie Li, "**MASQ: Multi-Agent Reinforcement Learning for Single Quadruped Robot Locomotion**". ICML 2025 @ NewInML.

12. Haonan Chen*, Bangjun Wang*, **Jingxiang Guo***, Tianrui Zhang, Yiwen Hou, Xuchuan Huang, Chenrui Tie, Lin Shao, "**World4Omni: A Zero-Shot Framework from Image Generation World Model to Robotic Manipulation**". ICML 2025 @ Building Physically Plausible World Models.

13. RoboTwin Challenge Organizers, **Jingxiang Guo** et al. (Volunteers), RoboTwin Challenge Participants, "**Benchmarking Generalizable Bimanual Manipulation: RoboTwin Dual-Arm Collaboration Challenge at CVPR 2025 MEIS Workshop**". Tech Report.

14. Qi Liu, **Jingxiang Guo**, Zhongjian Qiao, Pengbin Chen, Yanjie Li, "**Logarithmic Function Matters Policy Gradient Deep Reinforcement Learning**". Accepted to DAI 2024 (**Oral Presentation**).

15. Yiwen Hou, Yaoyu Cheng, Jiayu Luo, Zhixuan Xu, Chongkai Gao, **Jingxiang Guo**, Lin Shao, "**QuaMa: Extending Skill Learning with Delta Dynamics Modeling for Precise and Generalizable Quantitative Manipulation**". Under Review.

16. Qi Liu, Jianqi Gao, Dongjie Zhu, Zhongjian Qiao, **Jingxiang Guo**, Pengbin Chen, Yanjie Li, "**Multi-Agent Target Assignment and Path Finding for Intelligent Warehouse: A Cooperative Multi-Agent Deep Reinforcement Learning Perspective**". In Submission.

17. **Jingxiang Guo**, Jinxuan Zhu, Sixu Lin, Feng Shi, "**Momentum Prediction for Tennis Matches Based on Counter-Factual Analysis and Multi-LGBM**". Accepted to ICAACE 2024.

18. **Jingxiang Guo**, Jinxuan Zhu, Sixu Lin, Feng Shi, "**ECAPA-TDNN Embeddings for Speaker Recognition**". Accepted to AINIT 2024.

19. Kuntian Dai, **Jingxiang Guo**, Nengfeng Liu, Guanyu Hou, Jinbin Guo, Junkai Wang, Ruiquan Dong, "**Quick Reversing Device and Quick Track Reversing Device**". Invention Patent CN116000896B, Published 2025.

Awards

ICRA 2025 Best Paper Award on Robot Manipulation and Locomotion	**May 2025**
Best Robotics Paper Award at CoRL 2024 Workshop on MAPoDeL	**Nov 2024**
Best Paper Award at ICRA 2025 Workshop on Human-Centric Multilateral Teleoperation	**May 2025**
Best Demo Award Finalist at IROS 2025 Workshop on Contact and Impact-aware Manipulation	**Jun 2025**
National First Prize in RoboMaster University Championship (RMUC)	**Aug 2023**
National First Prize in China Intelligent Robots Innovation Competition	**Jul 2023**
Honorable Mention in Mathematical Contest in Modeling (MCM)	**Feb 2024**
Provincial First Prize in China Undergraduate Mathematical Contest in Modeling (CUMCM)	**Sep 2023**
Second Prize in National Embedded Chip Design Competition	**Jun 2023**
Second Prize in China Robotics Competition 2023	**Sep 2023**
Second Prize in Mechanical Engineering Innovation Competition 2023	**Sep 2023**
Second Prize in iCAN Innovation and Entrepreneurship Competition	**Nov 2023**
First Prize in RoboMaster University Championship 2023 Infantry Match	**Mar 2023**
First Prize in RoboMaster University Championship 2022 (National Final)	**Aug 2022**
First Prize in RoboMaster University Championship 2022 (Central China)	**Jun 2022**
First Prize in Hongli Cup Mathematical Modelling Competition	**Mar 2023**
Outstanding Student of the Year	**Nov 2022**
HITSZ Academic Scholarship	**Nov 2023**
Outstanding Member of the League	**May 2023**

Miscellaneous

Languages: English (IELTS 7.5, GRE 321), Chinese (Native)
Academic Service: Reviewer for ICRA 2025, ICRA 2026, RA-L, IROS 2025, CVPR 2025
Programming: Python, C/C++, MATLAB, HTML, CSS, JavaScript
Tools: SOLIDWORKS, ROS, PCB Design, STM32, Arduino, WebXR, LaTeX
Interests: Trekking and climbing